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找出 $n_1T^2\#n_2T^2, n_1T^2\#m_2P^2, m_1P^2\#m_2P^2$ 对应的曲面类型.

Solution: By definition $n_1T^2\#n_2T^2 = (n_1 + n_2)T^2, m_1P^2\#m_2P^2 = (m_1 + m_2)P^2$. Recall that $T^2\#P^2 = 3P^2$, so $n_1T^2\#m_2P^2 = (2n_1 + m_2)P^2$.

($K = 2P^2$ so $T^2\#P^2 \cong K\#P^2 \cong 3P^2$).

3

求证: 若 $f: \mathbb{S}^1 \rightarrow \mathbb{S}^n$ 不是满射, 那么 f 同伦于常值映射.

Proof: $\mathbb{S}^n \setminus \{x\} \cong \mathbb{R}^{n+1}$ is contractible, so f is null-homotopic.

4

设 $f, g: X \rightarrow \mathbb{S}^n$ 连续, 求证: 存在常数 $\varepsilon > 0$, 若 $\forall x \in X, d_S(f(x), g(x)) < \varepsilon$, 则 f, g 同伦.

Proof: Let $\varepsilon = \pi$, then $f(x) \neq -g(x), \forall x \in X$. Embed $\mathbb{S}^n$ into $\mathbb{R}^{n+1}$, and consider $H'(x, t) = (1 - t)f(x) + tg(x), t \in [0, 1]$. Since $f(x) \neq -g(x)$, we have $H'(x, t) \neq 0$ so $H: X \times [0, 1] \rightarrow \mathbb{S}^n$ is continuous where

$$H(x, t) = \frac{(1 - t)f(x) + tg(x)}{|(1 - t)f(x) + tg(x)|}$$

Then H is a homotopy between f, g .

5

证明 $n \geq 2$ 时 $\mathbb{S}^n$ 单连通.

证明: For any path $\gamma: [0, 1] \rightarrow \mathbb{S}^n$, since $n \geq 2$ and $[0, 1]$ is a 1-compact manifold, γ is homotopic to a polygonal curve $\tilde{\gamma}$. A polygonal curve has measure zero in an n -dimensional manifold so $\tilde{\gamma}$ is not surjective. Hence γ is homotopic to $\tilde{\gamma}$ which is null-homotopic, so $\mathbb{S}^n$ is simply connected.

6

设 $a, b \in \mathbb{C}$ 满足 $|a| > |b|$. 求证: $f(z) = \frac{az+b}{bz+\bar{a}}$ 定义的映射 $f: \mathbb{S}^1 \rightarrow \mathbb{S}^1$ 同伦与恒同映射.

Proof: We can find a homotopy $f_t(z) = \frac{az+tb}{tbz+\bar{a}}$ between f and $g(z) = \frac{a}{\bar{a}}z$ which is clearly homotopic to the identity.

Or one can check that f has exactly one zero in $\mathbb{D}$ so its degree is 1, and mappings $\mathbb{S}^1 \rightarrow \mathbb{S}^1$ of the same degree are homotopic.

7

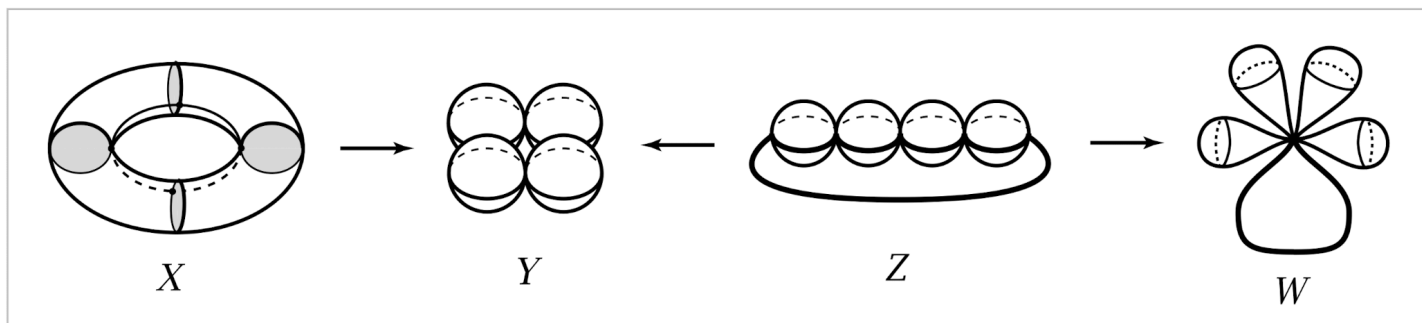
考虑

$$X = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1| = |z_2| = 1\} \cup \{(w, w) : |w| \leq 1\} \subset \mathbb{C}^2$$

证明 $X \cong \mathbb{S}^2 \vee \mathbb{S}^1$.

Proof: $A = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1| = |z_2| = 1\} \cong \mathbb{S}^1 \times \mathbb{S}^1$, $B = \{(w, w) : |w| \leq 1\} \cong \mathbb{D}^2$, $A \cap B \cong \mathbb{S}^1$. So X is a torus with a meridional disk (See Example 0.8&0.9 in Hatcher), which can be viewed as a CW-complex with two 0-cells (intersection points of the disk with the longitudinal circle), 1-cells of the circle and boundary of disk, and 2-cells of the torus and disk. Collapsing the disk B we obtain a sphere with two points identified, i.e. $X \cong \mathbb{S}^2/\mathbb{S}^0$.

Now consider a sphere with an arc a attached to two distinct points. By collapsing the arc a we obtain $\mathbb{S}^2/\mathbb{S}^0$, and by collapsing the arc b in $\mathbb{S}^2$ connecting the two endpoints of a , we obtain $\mathbb{S}^2 \vee \mathbb{S}^1$. Hence $X \cong \mathbb{S}^2/\mathbb{S}^0 \cong \mathbb{S}^2 \vee \mathbb{S}^1$.



8

把 $\mathbb{E}^3$ 中所有中心在 $(n, 0, 0)$ 半径为 $1/2$ 的球面并起来, 记作 X . 证明 X 单连通.

Proof: Since any path $f : \mathbb{S}^1 \rightarrow X$ has compact image, we only need to show that the finite union of such spheres is simply connected.

In fact, if X, Y are simply connected CW-complexes, then $X \vee Y$ is also simply connected, since van Kampen's Theorem implies $\pi_1(X \vee Y) = \pi_1(X) * \pi_1(Y)$. Note that $\mathbb{S}^2$ is simply connected, so the union of spheres is still simply connected.

9

求证: 若 $B \subset A \subset X$, B 是 A 的形变收缩核, A 是 X 的形变收缩核, 则 B 是 X 的形变收缩核.

Proof: Take (weak) retractions $X \xrightarrow{r_1} A \xrightarrow{r_2} B$ then $r_1|_A = 1_A, r_2|_B = 1_B$ so $r = r_2 \circ r_1 : X \rightarrow B$ satisfy $r|_B = r_B, 1|_X \cong r_1, 1|_A \cong r_2$ implies $r = r_2 \circ r_1 \cong 1_A \circ r_1 = r_1 \cong 1_X$. Hence r gives a weak deformation retraction $X \rightarrow B$.

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求证: 若 $B \subset A \subset X$, B 是 A 的强形变收缩核, A 是 X 的强形变收缩核, 则 B 是 X 的强形变收缩核.

Analogous to Problem 9.

11

连续映射 $f : X \rightarrow Y$ 的映射柱定义为 $L(f) = (X \times [0, 1]) \sqcup Y / \sim$ 其中 $(x, 1) \sim f(x)$. 求证: $L(f)$ 以 Y 为强形变收缩核.

Proof: Consider $f_t : L(f) \rightarrow Y$ given by $y \in Y \mapsto y$ and $(x, 1 - s) \in X \times [0, 1] \mapsto (x, 1 - st)$ then f_t is well-defined since $(x, 1 - s) \sim y \implies s = 0, f(x) = y$. Clearly $f : L(f) \times [0, 1] \rightarrow Y$ is continuous, and f_1

$= 1_{L(f)}, f_t|_Y = 1_Y$. Let $r = f_0$ then $\text{Im} r = Y$ and $f : r \cong 1_{L(f)} \text{ rel } Y$ so r gives a deformation retraction $L(f) \rightarrow Y$.

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连续映射 $f : X \rightarrow Y$ 的映射锥定义为 $C(f) = (X \times [0, 1]) \sqcup Y / \sim$ 其中 $(x, 1) \sim f(x), (x, 0) \sim (x', 0)$. 求证: $f_0 \cong f_1 \implies C(f_0) \cong C(f_1)$.

Proof: The mapping cone of $f : X \rightarrow Y$ is just an attaching space $C(f) = CX \sqcup_f Y$. We prove the following property: If (X_1, A) is a good pair and two attaching maps $f, g : A \rightarrow X_0$ are homotopic, then $X_0 \sqcup_f X_1 \cong X_0 \sqcup_g X_1 \text{ rel } X_0$. (See Hatcher Prop0.18).

Proof of lemma: If $F : A \times I \rightarrow X_0$ is a homotopy between $f = F_0, g = F_1$, consider $Z = X_0 \sqcup_F (X_1 \times I)$, then the deformation retraction $X_1 \times I \rightarrow X_1 \times \{0\} \cup A \times I$ induces a deformation retraction $Z \rightarrow X_0 \sqcup_f X_1$. Likewise, we have a deformation retraction $Z \rightarrow X_0 \sqcup_g X_1$ so they are equivalent under homotopy.